

When stock futures dominate price discovery

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Abstract

While the literature on price discovery attributes a weak role to single stock derivatives, this paper studies a large exchange where single stock futures dominate price discovery. The average information share of these derivatives rises to 55 percent when news arrives. The increase is larger for negative news where traders face restrictions on short-selling the spot. A partial equilibrium analysis predicts traders choose between the benefits of leverage and market liquidity, and these predictions are validated by an empirical panel analysis. The paper suggests that traders choose leverage to trade information and to preserve the efficiency of trading capital.

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1 Introduction

Among the many facets of price discovery in financial markets (Tookes, 2008), there is a puzzle about the role that single stock derivatives play. The literature, starting from Black (1975), predicts that traders prefer leveraged markets while trading information to enhance returns. That would suggest that the price of these derivatives, which are redundant securities except for their advantage of leverage, ought to dominate price discovery. However, almost all the empirical evidence on stock derivatives establishes the *spot* as dominating price discovery.

A recent focus on funding constraints of financial market participants (Brunnermeier and Pedersen, 2009; Acharya and Vishwanathan, 2011) introduces a new perspective on how traders can use leverage. If traders face funding constraints, they can use derivatives as lower margin securities rather than the spot, to improve the efficiency of limited trading capital. Higher funding constraints could thus result in traders using derivatives more and lead to higher average price discovery of the derivatives.

The paper identifies three gaps in the existing literature on price discovery. First, the measurement of price discovery in the existing literature is largely restricted to averages across time and securities. If stock derivatives are used to trade information, they will only dominate price discovery temporarily during periods of high information. An examination of periods surrounding news events would provide better insights into price discovery as a dynamic process (Chen and Gau, 2010; Brogaard *et al.*, 2012). Second, when trading is fragmented across different exchanges across different time zones and different microstructure, it is more difficult to assemble a comparable dataset to measure price discovery precisely. Third, the bulk of the price discovery literature focuses on the stock options market where the use of ‘implied price’ from the traded options price introduces estimation risk.

This paper focusses on the spot and the single stock futures (SSF) markets at the National Stock Exchange of India Ltd. (NSE), one of the biggest equity spot and derivatives exchanges of the world. The NSE has dominant market share on both the SSF and spot trading, thus avoiding the complexities associated with multiple trading venues. Both the spot and the SSF trade through an identical market design in the same time zone, which minimises the effect of microstructure noise. The observation of prices from a very liquid SSF market avoids the estimation risk associated with the implied price from options. These help to avoid the problems listed above.

The paper uses this setting to analyse variations in the price discovery of the SSF markets. A part of the analysis focusses on temporal variations in price discovery within the trading day in response to information arrivals, using periods such as the half hour at the start of the trading day following overnight news, or the half hour after a corporate announcement. Since there are restrictions on short-selling in the spot market which raises the costs of taking short positions in the stock, the analysis includes testing for any asymmetry of price discovery between periods of high negative and positive news to isolate the effect of funding

constraints on price discovery.

The second part focusses on the cross-sectional variation in price discovery, by considering the choices of a marginal trader who treats the leverage and the pre-trade liquidity of a stock as exogenously given. This partial equilibrium analysis concludes that the trader will choose the SSF if:

$$(\lambda - 1)r_f > c_f(\lambda Q) - c_s(\lambda Q)$$

where λ is the leverage of the SSF, r_f is the cost of borrowing for the trader, λQ is the transaction size on the SSF market, $c_f(\lambda Q)$ is the transactions cost faced by a market order on the futures market for an order size of λQ , and $c_s()$ is the same for the spot market. The expression shows that the trader will choose the SSF over the spot when the gains from leverage are larger than the differences in liquidity. This suggests some testable propositions: Price discovery is expected to take place to a greater extent on the SSF market for stocks where the liquidity difference, $[c_f(\lambda Q) - c_s(\lambda Q)]$, is negative and for stocks with greater leverage.

Price discovery is estimated using the Information Share (IS) measure (Hasbrouck, 1995) with prices from both the SSF and spot markets observed at a one-second resolution. The use of this very high frequency data from a single exchange gives the best estimation of IS. Concerns raised about the validity of the IS measure (Lehmann, 2002) are addressed by augmenting the inference as suggested by Yan and Zivot (2010). The analysis is carried out for 100 securities, observed for 104 days between March 2009 to August 2009. With a dataset of over 200 million records, this is one of the most detailed examinations of the price discovery process between the SSF and the spot.

There are three findings. First, the average SSF IS of 49 percent goes up to 55 percent during periods of high information. This is seen across all the stocks irrespective of their liquidity on either market, and demonstrates that traders prefer leverage to trade information. Second, the increase in price discovery of the SSF is higher for periods of large negative news which supports the hypothesis that traders facing higher funding constraints show a higher preference for the SSF. Third, a firm fixed effects regression of the daily stock level IS on leverage and the liquidity differences between the markets shows that the leverage coefficient is significant and positive, while the coefficient on the liquidity difference is significant and negative. Securities with higher leverage have higher IS of the SSF, as do securities with higher liquidity on the SSF market. This confirms the predictions of the partial equilibrium analysis.

The paper makes two contributions to the existing literature on price discovery. In contrast to previous studies, the paper shows that the share of SSF market can be as high as 49 percent. Higher IS values around high information periods provides evidence that traders choose derivatives market to trade information. It is also the first to present a partial equilibrium analysis of the choices of a marginal trader who takes leverage and liquidity as given. The findings depict price discovery as the outcome of the choice of traders as they respond to information arrivals and funding constraints, clarifying some of the puzzles about

the weak role of the SSF presented in the earlier literature.

The paper is organised as follows: Section 2 describes what is known so far about price discovery in the single stock derivatives markets, while Section 2.1 describes the microstructure setting of the NSE equity spot and SSF markets. The questions about price discovery in Section 3 has two broad parts: how average price discovery can be affected by both funding constraints and information in Section 3.1, and what drives the cross-sectional variation in price discovery in Section 3.2. The estimation methodology and testable propositions are presented in Section 4. Section 5 describes the data, followed by the results in Section 6. Section 8 concludes.

2 The research setting

Why do we expect traders to prefer the use of single stock derivatives to the underlying spot? The early hypothesis was that traders could use the leverage in these derivatives to enhance returns from information (Black, 1975). But, unlike the market index which has a constant flow of news and information, there is a lower frequency of news for a single stock. Thus, while informed traders are likely to use single stock derivatives in the event of information arrival, the steady flow of orders on the spot market are likely to be motivated by portfolio rebalancing or liquidity objectives.

This is consistent with the empirical literature where it is only for the index futures that a financial derivative unambiguously dominates price discovery.¹ However, traders can prefer to trade index derivatives for reasons other than leverage. Index derivatives offer the convenience of being a single contract using which to simultaneously trade all underlying index constituents.² Index futures do not suffer from the non-synchronous trading and spurious persistence of returns when trading involves all the underlying securities of the index (Miller *et al.*, 1994).

The role of leverage in price discovery is better analysed using single stock derivatives to avoid these confounding factors. But the empirical results about single stock derivatives have been a puzzle. Analysis based on daily data show that single stock options (SSO) prices lead the spot (Manaster and Rendleman, 1982), while later studies use high frequency data to show that the spot dominates the SSO.³ The literature on the role of single stock futures

¹A similar consensus does not hold for index options. For instance, Fleming *et al.* (1996) find that futures on the S & P 500 index lead options and both derivatives lead the spot S & P 500 index, but Chiang and Fong (2001) show that the index options lag the spot index.

²Instead of individual underlying index constituents, Hasbrouck (2003); Kurov and Lasser (2004); Tse *et al.* (2006) use prices from single index products such as Exchange Traded Funds (ETFs) and E-mini contracts and find that the index futures continue to dominate price discovery. This literature strengthens the case that traders prefer derivatives contracts to gain from their leverage.

³Stephan and Whaley (1990), Chan *et al.* (1993), Chan *et al.* (2002), Chakravarty *et al.* (2004). For example, Chakravarty *et al.* (2004) report that, on average, the SSO accounts for just 17% of price discovery,

(SSF) in price discovery is sparse but more consistently shows that the spot dominates price discovery.⁴

One potential explanation lies in the fact that this literature discusses the average behaviour of price discovery, which masks the variation in price discovery across time and across securities, while theory presents the use of derivatives as an outcome of sequential-trade models of trader's choices between derivatives and the underlying security.⁵ Several papers have researched the cross-sectional variation in price discovery, which is correlated with the market microstructure⁶ or features of the derivatives (such as moneyness and maturity of options).⁷ However, they do not explain why these leveraged products exhibit a low share in price discovery.

Another reason for traders to prefer leverage could be funding constraints. In a general equilibrium analysis, Brunnermeier and Pedersen (2009) show that traders who are funding constrained will shift to lower margin securities to preserve the efficiency of their trading capital during times of market stress. An analogy from this framework is that traders who are funding constrained would prefer derivatives because derivatives are low margin securities compared to the spot. It implies that such traders would prefer to trade the single stock derivatives to optimise their use of trading capital. At present, there are neither models of price discovery that incorporate funding constraints into how a trader chooses between the derivatives and the spot, nor is there empirical research about the role of these constraints in price discovery of derivatives.

In this paper, we exploit a unique market microstructure setting to explore this idea further. We focus on the price discovery of the single stock futures (SSF) because the SSF poses a simple analytical setting with a linear payoff⁸ that is directly proportional to the payoff of the spot by the amount of the leverage. This is unlike options where the size and effect of leverage is more complex to analyse. Next, we utilise the microstructure features of the equity markets at the National Stock Exchange in order to obtain improved measurement of price discovery.

and attributes the preference of informed traders for the spot market to the low liquidity in the options market.

⁴For example, Shastri *et al.* (2008) find that the SSF that trade on OneChicago contribute 24 percent to price discovery. Fung and Tse (2008) use minute-by-minute quotes data and find that the SSF on the Hong Kong Exchange only have a 30 percent share in price discovery. Kumar and Tse (2009) examine trades and quotes data, at the frequency of one minute, on 30 stocks listed on the NSE for a twelve month period in 2004 to show that SSF quotes are more informative than the spot, but that the traded spot price dominates price discovery.

⁵Biais and Hillion (1994); Easley *et al.* (1998); John *et al.* (2003).

⁶Fleming *et al.* (1996); Easley *et al.* (1998).

⁷Chakravarty *et al.* (2004); Anand and Chakravarty (2007)

⁸Other trading frictions can affect the payoff between the SSF and the spot. These frictions can be in the form of liquidity constraints that are symmetric, such as transactions taxes that apply equally to buyers and sellers, or are asymmetric, such as short-selling constraints on the spot market that affect sellers differently. These frictions would affect the size of the proportionality between the futures and the spot payoff, but not the linearity of the payoff itself.

2.1 A unique microstructure setting

The National Stock Exchange (NSE), which has traded equity since 1995, started trading derivatives in 1998 with index futures. But it was after the introduction of the single stock futures contracts that the derivatives traded volumes became significant.⁹ Out of the 1400 securities listed on the NSE, there are 223 securities that trade on the derivatives market during the period of the current paper. All the single stock derivatives contracts have maturities from one to three months, but most of the trading is concentrated in the one month contract.

NSE trades both the spot and the derivatives simultaneously¹⁰ on anonymous, electronic limit order-book systems. Orders that arrive during trading hours, which are between 9:55 am and 3:30 pm¹¹ are matched on a price-time priority. Till 2010, all equity derivatives trading in India has been concentrated on the NSE,¹² while trading on the equity spot has been spread across the two primary equity exchanges.¹³

The settlement of equity spot trades are done on a $T + 2$ basis, which is a shorter period than in most international markets. The shorter time-frame imposes a greater pressure on the market participants to make delivery of funds or securities. Margin payments for the derivatives are made upfront, with the mark-to-market profit/loss being settled on a $T + 1$ basis. However, because of the leverage, the size of the settlement in the SSF is much smaller.

Other features of the equity settlement process which can affect the choice of traders is that (a) short-selling is not permitted on the spot market, and (b) the SSF are cash-settled. While all short positions on the spot must deliver the underlying equity on a $T + 2$ basis, the short positions on the SSF need only deliver funds. Since there is no well developed market or exchange mechanism for lending and borrowing shares, short-sellers tend to face higher funding constraints while selling on the spot.

In this environment, the traded volumes on the SSF market has been consistently higher than the spot market, unlike in most other global SSF exchanges. Table 1 shows the list of the top global exchanges with significant trading in SSF between 2008 and 2011, where the NSE is consistently ranked third highest by number of contracts traded.

In marked contrast to the high volumes of the SSF, there is very little trading on the SSO at the NSE, which is also unlike what is observed on the better known, global equity derivatives

⁹The SSF trading replaced a traditional market mechanism called *badla* which gave traders access to leverage through forward trading (Berkman and Eleswarapu, 1998). The familiarity of the brokers with the *badla* system facilitated an easy transition into trading the SSF.

¹⁰In India, exchanges are differentiated by asset classes (equity, fixed income, commodities) rather than spot and derivatives. This is unlike the well-studied markets such as in the U.S., where the spot and the derivatives trade on different exchanges.

¹¹This has been changed to 9 am to 3:30 pm since January 1, 2010.

¹²Indian securities market, a review (ISMR) - 2009 http://www.nseindia.com/content/us/ismr_full12009.pdf.

¹³These are the NSE and the Bombay Stock Exchange (BSE).

Table 1 Top derivatives exchanges for single stock futures contracts

The table presents the number of single stock futures (SSF) contracts traded in the top exchanges of the world, ranked by the largest number of SSF contracts traded between 2008 and 2011.

Rank	Exchange	(Million)			
		2008	2009	2010	2011
1	RTS Russia		249	227	355
2	NYSE Liffe Europe	125	197	289	246
5	NSE India	226	161	176	161
4	ASX Australia	69	142	153	168
3	EUREX	130	117	198	171
6	Johannesburg Stock Exchange	420	89	79	51
7	KOREA Exchange	12	37	45	60

Source: World Federation of Exchanges, <http://www.world-exchanges.org>

exchanges. During the sample period of this paper, the traded volumes on the SSO market constituted only 7.56 percent of the total volumes traded on all single stock derivatives. This has two implications for the analysis in this paper: 1) There is insufficient data to enable us to carry out an analysis of price discovery of the SSO market on the NSE, and 2) It simplifies the analysis since price discovery will take place on either the spot or the SSF markets at the exchange.

Who trades?

The trading on the NSE equity markets is dominated by relatively small financial firms, rather than the large financial firms such as banks, asset management, insurance and pension firms that dominate global markets. Only around 15 percent of the trades are from the traditional institutional firms such as fund houses and asset management companies.¹⁴ The remaining trades come from non-institutional sources which includes retail, proprietary trades by brokerage firms, high-frequency traders, or hedge fund-like entities.

This is the outcome of regulatory restrictions. Banks are constrained on how they can fund trading activity in the equity spot markets.¹⁵ They are not permitted to fund activity in equity derivatives, particularly arbitrage.¹⁶ Both insurance firms¹⁷ and pension funds¹⁸ face similar restrictions on investments in equity and in derivatives. Capital controls mean that

¹⁴In Table 1.5 of *Institutional, Retail & Proprietary Investors - Turnover Analysis*, NSE (Jun 2010).

¹⁵From the Reserve Bank of India (RBI) Master Circular on Exposure Norms, DBOD No. Dir. BC. 7/13.03.00/2011-12 dated July 1, 2011, <http://rbidocs.rbi.org.in/rdocs/notification/PDFs/68MC010712EF.pdf>

¹⁶From the RBI draft comprehensive guidelines on derivatives, http://www.rbi.org.in/Scripts/bs_viewcontent.aspx?Id=457

¹⁷From the website of the insurance sector regulator at <http://www.irdaIndia.org/regulations.htm>

¹⁸From the website of the New Pension Systems regulator at http://pfrda.org.in/writereaddata/linkimages/investment%20guidelines%2030_04_092418358081.pdf

foreign institutional investors (FIIs) also have regulatory and operational restrictions on how they can participate in the equity markets (Shah *et al.*, 2008).

In the sample period analysed in this paper, there were no designated market makers for either the spot or the SSF, although there are no regulatory restrictions on traders to place limit orders on both sides of the market simultaneously. The amount of capital that is deployed into trading activity is relatively small compared to the larger, developed economies. Data from the NSE shows that the top 10 member firms account for 25 percent of the annual turnover.¹⁹ Independent data from the balance sheets of listed securities firms shows that the top 10 firms had USD 160 million of equity capital. When compared against an annual turnover²⁰ of USD 6.8 trillion on the NSE equity markets, these are small numbers for equity capital and suggests significant funding constraints for trading firms on this exchange.

3 Questions

Within the setting described above, the paper asks the following questions:

1. Is there temporal variation in the price discovery of the SSF, and how does it vary with information arrival and trading constraints?
2. Is there cross-sectional variation in price discovery, and how does it vary with leverage and market liquidity of a security?

3.1 Temporal variation in price discovery

Two factors can influence a trader’s use of leveraged products: the arrival of information and the funding constraints of traders. Even though they affect price discovery in the same direction (higher price discovery on the SSF), they differ in the nature of their influence on trader choices. First, information is a temporary influence on the trader’s choice, while funding constraints are time invariant. Second, information affects *all traders* when it arrives, while funding constraints can vary *across traders*. Hence, intra-day price discovery would be caused by information arrival and not funding constraints, but funding constraints will affect the average level of price discovery in the market. So far, empirical research has focussed mostly on the average levels of price discovery in the market.

A small recent literature has used high-frequency data to detect intra-day changes in price discovery in response to information. Muravyev *et al.* (2012) analyse the price discovery of options markets during high information periods, and find that these prices do not contain

¹⁹From the NSE website at http://www.nseindia.com/products/content/equities/equities/historical_topbrokersyearwise.htm

²⁰This is the sum of the annual turnover on the spot market and the futures and options contracts on equity, including both the index and the single stock products.

significant information about future stock prices. They estimate an average information share of 6.25 percent for the options market. Dong and Sinha (2012) analyse the price discovery surrounding news events using SSO prices data at five-minutes frequency. They find that the information share of the option markets rises significantly, from 10 percent during the non-information periods to 27 percent during the high information periods.

We identify time-series variation in price discovery in a manner similar to Muravyev *et al.* (2012). The price discovery measures are estimated over smaller time intervals within the trading day which are identified as periods of high information. However, unlike Muravyev *et al.* (2012) who identify such periods using violations of put-call parity in SSO prices, our identification approach uses periods of high information. One such period is during the start of the market, when traders respond to accumulated overnight news and information (Greene and Watts, 1996; Kalev *et al.*, 2004). Another such period is the time around corporate announcements (Cao *et al.*, 2005; Roll *et al.*, 2010). Once these periods are identified, they can be used to test the following hypothesis about the process of price discovery of the SSF markets:

HYPOTHESIS H¹: The role of the SSF in price discovery, during periods of high information within the trading day, is higher than during other periods.

H^1 tests whether the price discovery of the SSF rises during information arrivals, irrespective of the direction of the news. If transactions costs are the same for taking long or short positions and if traders face no funding constraints, then the decision about whether to trade SSF or the spot would be the same regardless of whether the expected price move is positive or negative. However, if traders are funding constrained, then their choice of trading venue would be influenced by differential costs in the two markets for long and short positions. On the NSE, short-selling restrictions on the spot make the transactions costs of using the spot higher when the trader wants to short the stock upon the arrival of negative news. Thus, funding constrained traders ought to have a *stronger preference* for SSF in response to negative news compared to positive news. This leads to the following hypothesis:

HYPOTHESIS H²: The role of the SSF in price discovery during periods of high information is higher when the news is *negative* compared to when the news is positive.

3.2 Cross-sectional variation in price discovery

In addition to temporal variation in price discovery, there can also be cross-sectional variation at a given point in time. Since leverage is a key factor identified behind the use of derivatives and varies across stocks, we focus on cross-sectional variation in price discovery by leverage. Among the rich base of research on cross-sectional variation in price discovery of the SSO markets, there are few papers that explicitly include leverage as an explanatory variable. Instead, the volatility of the stock is used as a proxy for leverage. For exchange-traded derivatives, leverage is determined by the margin rules of the exchange clearing house

(Kupiec, 1989, 1998) which are, in turn, based on volatility. The higher the volatility, the higher the margins charged, and the lower the leverage. This is consistent with Chakravarty *et al.* (2004) who show that price discovery of the SSO markets tends to be lower when volatility is higher.

Chakravarty *et al.* (2004) also show that price discovery by the SSO varies by both the liquidity in the market as well as with product features such as the strike and the maturity.²¹ Shastri *et al.* (2008) find that price discovery on the U.S. SSF markets tends to be higher on days when they are more liquid, even though the spot market continues to dominate.

On the electronic limit order book market, traders can observe the available liquidity in the market *before* they place their order through the set of quantities and prices at which they can buy or sell shares. This *pre-trade liquidity* of the stock (henceforth, referred to simply as “liquidity”) on alternative trading venues may influence where to trade. Any trader can observe the following features about the markets before choosing where to trade: λ , the leverage available in the SSF market; $c_{\text{SSF}}(Q)$ which is the (pre-trade) liquidity on the SSF market, measured as the average cost of trading a market order of size Q on the SSF market; and $c_s(Q)$ is, similarly, the (pre-trade) liquidity on the spot market.

We build a simple framework to understand how a trader chooses between alternative trading venues, when a given security trades simultaneously on the SSF and the spot market. Our analysis focuses on the decision of a marginal trader who can treat liquidity and leverage as exogenously given. We assume that this trader acts in order to maximise the ratio of expected returns to the risk taken. While we recognise that liquidity and leverage themselves are shaped by the choices of traders, we restrict ourselves to a partial equilibrium analysis. We leave it to the empirical analysis to show the usefulness, or lack thereof, of this modeling simplification.

Appendix A has the proof of the result that the trader will prefer to trade on the SSF when:

$$(\lambda - 1)r_f > [c_{\text{SSF}}(Q) - c_s(Q)] \quad (1)$$

Equation 1 captures the trade-off between the leverage and the liquidity of the spot and the SSF market in our simplified model. The term $(\lambda - 1)$ is the gain from leverage, calculated as the difference in the funding costs to the trader for a market order of the same size Q on both the spot and the SSF market. This will always be positive for the SSF. The term $[c_{\text{SSF}}(Q) - c_s(Q)]$ is the liquidity difference measured as the difference in the transaction cost of trading a market order of size Q on the SSF, $c_{\text{SSF}}(Q)$, and the spot market, $c_s(Q)$. This difference could either be positive or negative, depending on market conditions at the time that the trader chooses.

The trade-off shows that a marginal trader will prefer the SSF as long as the gain from leverage is greater than the liquidity difference. This is similar to the finding by Easley *et al.* (1998) that if the leverage effect of options is large enough, or if the liquidity in the spot

²¹In fact, papers such as Easley *et al.* (1998), Pan and Poteshman (2006) and Anand and Chakravarty (2007) show that *traded volumes* on the SSO market can predict spot prices

market is poor, then at least some traders will use the options. One implication of Equation 1 is that neither the expected returns $E(r_s)$ nor the risk of the security σ_s matters, even though the model sets up the return and risk of the spot as different from the SSF. It is only the microstructure that influences the choice of trading venue.

This analysis suggests two testable hypotheses about cross-sectional variation in the price discovery of the SSF:

HYPOTHESIS H³: There is a greater role of the SSF in price discovery for securities where the liquidity difference ($[c_{\text{SSF}}(Q) - c_s(Q)]$) is relatively more negative, *holding the leverage constant*.

HYPOTHESIS H⁴: The price discovery of the SSF is higher for securities where the leverage is relatively higher, *holding the liquidity difference constant*.

4 Estimation

We discuss two issues related to estimation. The first is about the interpretation of the information share measure. The second concerns testing the hypotheses, H¹ ... H⁴, outlined in the previous section.

4.1 Interpretation of the information share (IS) measure

There are two measures that are typically used for price discovery across multiple markets. These are the Information Share (IS) proposed by Hasbrouck (1995), and the Component Share (CS) which is based on the permanent and transitory decomposition of Gonzalo and Granger (1995). There has been a significant debate on the correct interpretation of each measure. Lehmann (2002) points out that problems of interpretation are inevitable when the measures are derived from a (reduced form) vector error correction model, but the inference about the importance of any single market to price discovery depends upon the parameters of the underlying structural model.

Yan and Zivot (2010) attempt to resolve the confusion by proposing a structural cointegration model for the price changes in multiple markets. They re-interpret the IS and the CS in terms of the underlying structural innovations: permanent, information-related innovations (η_t^P) due to the arrival of news and transitory, and non-information-related innovations (η_t^T) due to trading frictions. They show that both measures adjust for the relative avoidance of transitory shocks, but that only the IS captures the relative informativeness of a given market. They suggest the joint use of the CS and IS values to separate the effects of η_t^P and η_t^T , and to achieve sound interpretation about which market leads in price discovery.

We build on this suggestion and combine the IS and CS estimates to create the IS-CS ratio as:

$$\frac{|IS_1 \times CS_2|}{|IS_2 \times CS_1|} = \frac{|d_{0,1}^P|}{|d_{0,2}^P|}$$

where IS_1 is the information share for Market 1, CS_2 is the component share for Market 2. $d_{0,1}^P$ captures the contemporaneous response of Market 1 to a permanent shock, while $d_{0,2}^P$ captures the same for Market 2. The IS-CS ratio captures the contemporaneous response of each market to permanent shocks. A value that is greater than one implies that Market 1 reacts to permanent shocks more strongly than Market 2.²² Thus, the earlier literature would infer that “Market 1 leads Market 2” from the result that “ $IS_1 > 0.5$ ”. When the same result is used with the IS-CS ratio, it can support one of three possible interpretation:

1. Market 1 leads Market 2 if the ($IS_1 > 0.5$) *and* the (IS-CS ratio > 1).
2. Market 2 has a stronger response to transitory shocks due to which IS_1 is high, and it cannot be inferred that Market 1 leads Market 2, if ($IS_1 > 0.5$) and the (IS-CS ratio < 1).
3. Market 2 leads Market 1 if the ($IS_1 < 0.5$) *and* the (IS-CS ratio < 1).

The use of IS-CS ratio jointly with the IS_{SSF} helps to strengthen the interpretation that the estimated values of IS for a market represent stronger response of that market’s price to information and not to noise.²³

4.2 Testing the hypotheses

In the following, we discuss the approach used to test each of the four hypotheses, $H^1, \dots, H^4$, presented in Section 3 about variations in price discovery of the SSF.

H^1 : *Test the response to information arrival.*

1. Identify periods when there is high information in the market, H, within any given trading day t .
2. Estimate $IS_{SSF,i,t,H}$ for each security i in the sample, for each day t , separately during periods of high information H identified above, and $IS_{SSF,i,t,L}$ during periods of low information L.
3. Estimate a fixed effects panel regression of the form:

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1 \text{HIGH-INF}_j + \epsilon_{i,t,j} \quad \forall j = H, L$$

where $\text{HIGH-INF}_j = 1$ when $j = H$ during the trading day, and 0 when $j = L$.

²²A detailed description of Yan and Zivot (2010) findings as well as the derivation of the IS-CS ratio can be found in Section B of the Appendix. This section also contains an example using one day of data for a firm in the dataset.

²³A very recent paper endorses this approach (Putnins, 2013).

4. Test the null:

$$H_0^1 : \beta_1 = 0; \quad H_A^1 : \beta_1 > 0$$

If H_0^1 is rejected, we conclude that price discovery of the SSF market rises in response to periods of high information.

H² : *Test an asymmetric response to positive and negative information.*

1. Identify periods when there is high information in the market, H, within any given trading day t .
2. Estimate $IS_{SSF,i,t,H}$ for each security i in the sample, for each day t , separately during periods of high information H identified above, and $IS_{SSF,i,t,L}$ during periods of low information L.
3. Separate the high information periods H identified above into periods of positive news, H^+ , and negative news, H^- .
4. Estimate a fixed effects panel regression of the form:

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1^+ HIGH-INF_j^+ + \beta_1^- HIGH-INF_j^- + \epsilon_{i,t,j} \quad \forall j = H, L$$

where $HIGH-INF_j^+ = 1$ when $j = H$ and positive and $HIGH-INF_j^- = 1$ when $j = H$ and negative during the trading day, and 0 when $j = L$.

5. Test the null:

$$H_0^2 : \beta_1^+ - \beta_1^- = 0; \quad H_A^2 : \beta_1^+ - \beta_1^- < 0$$

If H_0^2 can be rejected, we conclude that funding constraints drive traders to trade the SSF more during periods of high, negative information.

H³, H⁴ : *Test the influence of security-specific leverage and market liquidity.*

1. Estimate $IS_{SSF,i,t}$ and leverage $\lambda_{i,t}$ for each stock i in the sample, using the price data available at one second frequency for each trading day, t .
2. Measure the time series of daily liquidity difference for each stock i , $(IC_{SSF,i,t} - IC_{S,i,t})$ from the limit order book.
3. Estimate the cross-sectional relationship between price discovery, leverage and the liquidity difference between the SSF and the spot market using a fixed effects panel regression of the form:

$$IS_{SSF,i,t} = \alpha_i + \beta_2(IC_{SSF,i,t} - IC_{S,i,t}) + \beta_3\lambda_{i,t} + \epsilon_{i,t}$$

where the intercept term, α_i captures all security specific information other than the leverage and the liquidity difference.

4. Test the null:

$$\begin{aligned} H_0^3 : \beta_2 &= 0; & H_A^3 : \beta_2 < 0 \\ H_0^4 : \beta_3 &= 0; & H_A^4 : \beta_3 > 0 \end{aligned}$$

If the associated nulls cannot be rejected using a one-tailed test, we conclude that leverage and the liquidity difference can explain cross-sectional variation in price discovery of the SSF.

5 Data description

The dataset used in the analysis comprises observations of the limit order book for both the NSE equity spot and SSF markets, at a frequency of one second. This allows us to estimate intra-day values of price discovery, liquidity and leverage for a given security. The use of very high frequency data mitigates the problem of high correlations that may arise due to temporal aggregation at lower frequencies (Hasbrouck, 1995). We focus on the 100 firms which make up the CNX-100 index. These securities account for approximately 82 percent of the total SSF market volumes and 51 percent of the total traded volumes on the spot market.²⁴ However, since three out of the 100 securities have no trades on the SSF during the sample period, we restrict our analysis on the remaining 97 securities.

The period that we analyse is from March 2009 to August 2009, a total of 104 trading days. Midquote prices of the near-month futures contracts and spot are used in the analysis, with the rollover to the next month futures contract prices taking place two days prior to expiration. In all, the sample has 10,088 stock days of data for both spot and the SSF markets. With 20,000 seconds per day, the overall data has over 200 million records.

In the following sections, we present the sample characteristics for the input variables used in testing hypotheses $H^1 \dots H^4$.

5.1 Estimates of IS

Sample estimates of IS are presented in Table 2, along with the IS-CS ratio. The overall sample average IS_{SSF} is 49 percent. The comparable IS_s is 51 percent. These estimates do not differ at the 95% confidence level. The average sample IS-CS ratio is 1.01, from which we infer that the SSF market reacts as much to the permanent shocks as does the spot market.²⁵

²⁴Source: NSE website, <http://www.nse-india.com>

²⁵The IS-CS ratio is based on the assumption of uncorrelated contemporaneous residuals. In the sample, the average correlation is 0.05, which is insignificantly different from zero, and supports the assumption of uncorrelated residuals.

Table 2 Summary statistics of overall sample estimates of IS_{SSF} and IS_s

The table summarises the price discovery estimates for both the SSF (IS_{SSF}) and the spot (IS_s) markets. These are calculated daily for each stock and averaged across all traded days for each security. The table also reports the value of IS-CS ratio to augment inference.

	Min	Mean	Median	Max	Variance
IS_{SSF}	0.05	0.49	0.56	0.75	0.20
IS_s	0.95	0.51	0.44	0.25	0.18
IS-CS ratio	0.30	1.01	1.06	1.74	0.31

We find that all the securities in the sample meet the first two conditions specified in Section 4.1 for sound inference: (1) when $IS_{SSF} > 0.5$, IS-CS ratio is greater than one and (2) when $IS_{SSF} < 0.5$, IS-CS ratio remains less than one. This implies that the high estimates of IS for a market can be correctly interpreted as indicating that market’s response to permanent shocks, unlike earlier studies on the price discovery of single stock derivatives which report only the IS estimate.

5.2 Identifying high information periods

In order to test for the behaviour of price discovery during high information periods, we use two sets of periods that have previously identified in the literature as being high information periods in financial markets. These are:

High market-wide information at the start of the trading day.

It is well documented that intraday market volatility follows a U-shaped pattern, where the volatility peak at the start of the trading day is attributed to the inflow of overnight news into the market. Thomas (2010) tests for structural breaks in the intraday market volatility using the Nifty index, and finds that the high volatility at the start of the trading day persists for half an hour on average. In this paper, we use the period from 09:55 am to 10:30 am on each day in the sample as a period of high information. We use the period between 12:00 pm to 1:00 pm as the comparative quiet period.

High security-specific information during corporate announcements.

Another well-documented occurrence of information is the period immediately following corporate earnings announcements (Ball and Kothari, 1991). These are security specific announcements, that can arrive randomly within a trading day. The date and time-stamps of earnings announcements for the securities in our sample are hand-collected from archives at the NSE website, and cross-validated with information about the same firms from the Bombay Stock Exchange website. We find that there are two periods in the sample during which earnings are announced: a) Results for the last quarter of the financial year (FY) 2009,²⁶ announced during the months of April and May 2009, and b) results for the first quarter of (FY) 2010, announced during July and August 2009. The first half hour immediately after

²⁶The Indian financial year is from Apr 1 to Mar 31.

the earnings announcements is used as the period of security specific high information. There are 158 such instances in the sample.

5.3 Liquidity difference between the SSF and spot markets

Every trader sees the best five available quotes on both sides of the limit order book which allows him to calculate the *impact cost*, $IC(Q)$, of a market order of any size Q before placing the order. $IC(Q)$ is calculated as:

$$IC(Q) = 100 \times \frac{P_Q - P_{MQ}}{P_{MQ}} \quad (2)$$

where P_Q is the execution price calculated for a market order of Q and P_{MQ} is the mid-quote price. For a liquid security, the IC_Q for a very large Q will be small (i.e., the execution price will be close to the mid-quote price). $IC(Q)$ can be calculated for many values of Q , given the visible part of the LOB. In this paper, we set the order size $Q = \$5000$ which is the smallest size of trade permitted on the SSF market.

Traditional measures of liquidity such as traded volumes and the bid-ask spread can also be readily observed from the LOB, but these are not useful to directly measure the transactions costs across the markets required in Equation 1. For example, the bid-ask spread is typically reported for non-comparable transaction sizes across the spot and SSF markets.²⁷ Another reason to look beyond the spread is that informed traders are more likely to desire rapid execution of larger order sizes, beyond the quantity at the spread. Several papers in the literature attempt to model liquidity as a function of Q but there is no consensus on the functional form.²⁸ By measuring $IC(Q)$ directly, we avoid these issues.

Table 3 presents the summary statistics for the impact cost values in comparison to the traditional measures of traded volumes (as the number of shares traded) as well as the average bid-ask spread (as a percentage of the mid-quote price). The table shows that the SSF market has higher post-trade liquidity (traded volumes) on average, but the spot market has better pre-trade liquidity (bid-ask spread and impact cost).

²⁷For instance, the average spot equity market trade on the NSE is USD 500 on average, while that on the SSF is USD 5000. The bid-ask on the SSF market is for quantities that are 10 times larger than those on the spot market, on average.

²⁸Papers that propose non-linear functions of liquidity costs as a function of trade size Q include Potters and Bouchaud (2003), Almgren *et al.* (2005), Huang and Ting (2008), and Tayal and Thomas (2012). Others that find no significant deviation from linearity include Engle and Lange (2001) and Breen *et al.* (2002). Rosu (2009) shows that the impact cost increases in Q as either a quadratic or an exponential function or a mixture of both.

Table 3 Sample liquidity characteristics

The table present averages values of traditional liquidity measures along with the impact cost (IC) for all securities in the sample as an illustration. The IC for each security has been calculated for market orders of \$5000. The reported values of the bid-ask spread and the IC have been calculated as the median value for each security, which is then averaged for the full sample.

	SSF	Spot
Volumes ('000 shares)	6,337	5,267
Bid-ask spread (%)	0.23	0.05
Impact cost, IC (%)	0.24	0.13

5.4 Leverage

Leverage for security i on date t , $\lambda_{i,t}$, is estimated from the margin requirements imposed by the exchange on the SSF contracts. This, in turn, is based on the volatility of the security, $\sigma_{i,t}$.²⁹ The methodology used in this paper to estimate $\lambda_{i,t}$ for security i for each trading day t , is described in detail in Section C of the Appendix.

Figure 1 Cross-sectional variation in the leverage for the sample

The graph shows the cross-sectional variation in the leverage for the securities in the sample vis-a-vis the annualized volatility. The estimated leverage values range from $2.5\times$ to $8\times$. There are very few securities with leverage at 8, which is the maximum permitted leverage for the SSF at the NSE.

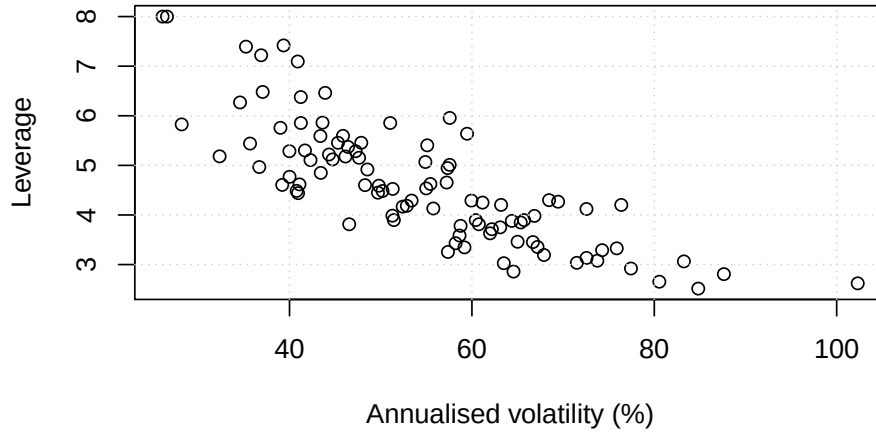


Figure 1 illustrates the variation in the estimated leverage across the securities in the sample. The values of the average leverage range from 2.5 to 8 percent with very few observations

²⁹The amount of leverage calculated at the NSE is the sum of the Value at Risk (VaR) and an Exposure Margin (EM). Currently, the minimum VaR and EM imposed by the NSE is 7.5% and 5%, respectively. This implies that the minimum margin permitted is 12.5%, and the maximum leverage permitted is thus 8 times. The exchange limits the leverage at 8.

Source: <http://www.nseindia.com/products/content/derivatives/equities/margins.htm>

Table 4 Sample statistics of liquidity and leverage by quartiles

Sample statistics for both SSF and spot are presented for quartiles based on:

1. Liquidity $(c_{\text{SSF},i} + c_{\text{S},i})/2$
2. Leverage $(\hat{\lambda}_i)$

in that order. For each quartile, the table presents the median value of the liquidity for the securities in that quartile, and the minimum, average, and maximum leverage for the same.

Quartiles	Q1 (Most)		Q2		Q3		Q4 (Least)	
	SSF	Spot	SSF	Spot	SSF	Spot	SSF	Spot
By Liquidity								
IC_μ (%)	0.04	0.04	0.07	0.06	0.13	0.11	0.74	0.29
$\hat{\lambda}_\mu$	4.44		4.40		4.57		5.07	
By Leverage								
IC_μ (%)	0.32	0.17	0.12	0.10	0.21	0.14	0.10	0.08
$\hat{\lambda}_\mu$	6.15		4.86		4.07		3.17	

that are close to, or at, the maximum leverage permitted by the NSE. Thus, the limit is unlikely to be a binding constraint for the analysis in this paper.

5.5 Correlation between liquidity and leverage

Lastly we examine the cross-sectional variation in sample liquidity and leverage in Table 4, using sample means for quartiles by liquidity and leverage. We find little evidence of a consistent pattern in leverage across the liquidity quartiles, or in liquidity across leverage quartiles, and infer that neither of these variables present a problem of multicollinearity in the regressions proposed in Section 4.2.

6 Results

We present the analysis of the hypotheses, $H_1, \dots, H_4$ in two ways. The first is a non-parametric approach by examining the average IS_{SSF} for the full sample and by quartiles. The second is a parametric analysis based on estimations from fixed effects panel regressions.

6.1 Temporal variation in price discovery

The summary statistics for the IS_{SSF} values in Table 2 show that the average IS_{SSF} estimated for the NSE SSF market is higher than those reported in the previous literature.³⁰ Thus,

³⁰For example, Chakravarty *et al.* (2004) report that on an average, single stock options market accounts for just 17% of price discovery. Shastri *et al.* (2008) find that SSF contribute 24% in price discovery using

the NSE SSF markets have both high price discovery as well as high liquidity, compared to what has been previously reported in the literature.

Table 5 Summary statistics of IS_{SSF} for high and low information periods

The table shows values of IS_{SSF} and IS_s during periods of high information and periods of low information, as defined in Section 3. The IS_{SSF} presented is the average of the upper and lower bound values estimated for the securities in the sample both during the first half hour of trading and the first half hour after the corporate earnings announcement. The value of IS-CS ratio helps to clarify inference.

Period of	Min	Mean	Median	Max	Variance
High Information					
IS_{SSF}	0.13	0.56	0.61	0.85	0.17
IS_s	0.15	0.44	0.39	0.87	0.17
IS-CS ratio	0.36	1.15	1.19	2.06	0.33
Low Information					
IS_{SSF}	0.10	0.49	0.55	0.73	0.16
IS_s	0.27	0.51	0.45	0.90	0.16
IS-CS ratio	0.42	1.00	1.03	1.70	0.27

Hypothesis H^1 tests for changes in price discovery of the SSF market during periods of high information flow. Table 5 presents the average values of both IS_{SSF} and IS_s for high information (in the first panel) and low information periods (in the second panel). A comparison of the sample median shows that even during low information periods, the SSF market dominates the spot with a value of 55 percent on average. This increases to a further 61 percent during the high information periods, which is a 9 percent increase. To the best of our knowledge, this is the first finding of a dominant SSF market in the price discovery process.

We further test the response to high information using the following firm fixed-effects regression:

Model 1 : $IS_{SSF,i,t,H/L}$ on a high information dummy variable, $HIGH-INF_j$

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1 HIGH-INF_j + \epsilon_{i,t,j}$$

where j = takes value 1 for a high information period, H and 0 for a low information period, L.

The results presented in Table 6 show that in Model 1, $\hat{\beta}_1$ is positive and strongly significant. The average price discovery of the SSF goes up by nearly 6 percentage points during periods of high information, validating H_A^2 that traders use the SSF market more during information arrival.

This increase in IS_{SSF} is similar to that reported by Dong and Sinha (2012), who find that the information share of the SSO market increases around news events. However, they report a

data on SSF that trade on OneChicago. Fung and Tse (2008) conclude that the SSF have only a one-third share in price discovery based on minute-by-minute quotes data from the HongKong Exchange.

Table 6 Fixed effects panel regressions of intraday variation in IS_{SSF} during high information periods

The table presents regression results for a fixed effects model, Model 1:

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1 HIGH-INFO_{t,j} + \epsilon_{i,t,j}$$

where $HIGH-INFO_{t,j} = 1$ for $j = H$ for high information periods during the trading day t and 0 for all other periods.

A second fixed effects model, Model 2, additionally tests for any effects of the benefits of short selling to explain the variation in IS_{SSF} during high information periods.

Model 2:

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1^+ HIGH-INFO_{t,j}^+ + \beta_1^- HIGH-INFO_{t,j}^- + \epsilon_{i,t,j}$$

where $HIGH-INFO_{t,j}^+ = 1$ for $j = H$ when the index returns are positive during the high information periods on trading day t , zero otherwise. Similarly, $HIGH-INFO_{t,j}^- = 1$ for $j = H$ when the index returns are negative during the high information periods, zero otherwise.

Values marked in boldface indicate that the coefficients are significant at 5 percent, based on clustered standard errors.

	Model 1			Model 2		
	Estimate	Std. Error	% of obs.	Estimate	Std. Error	% of obs
HIGH-INFO	0.061	0.005	50.86			
HIGH-INFO ⁺				0.055	0.006	23.36
HIGH-INFO ⁻				0.069	0.006	24.10
Adjusted R ²	0.20			0.20		

much lower value of 27 percent for the average IS_{SSO} compared to our estimate of 56 percent for IS_{SSF} around news and information. This might reflect a combination of the effect of high information, and higher funding constraints of the Indian trader.

We next test for an asymmetry of the effect between high positive and negative information. Since there are restrictions on short-selling in the spot market, there will be a higher cost of trading the spot when information is negative. In order to incorporate this aspect into the test for asymmetry in price discovery, we modify the fixed-effect regression in Model 1 by replacing the single $HIGH-INFO$ dummy with two dummies, $HIGH-INFO_j^+$ and $HIGH-INFO_j^-$, which are defined as follows:

$$HIGH-INFO_j^+ = \begin{cases} 1 & \text{if } r_{m,t,j} > 0; \\ 0 & \text{otherwise} \end{cases}$$

and

$$HIGH-INFO_j^- = \begin{cases} 1 & \text{if } r_{m,t,j} < 0; \\ 0 & \text{otherwise} \end{cases}$$

where $r_{m,t,j}$ denotes returns on the market index from the previous closing price to the price at the end of the high information period of the trading day. We estimate the following firm fixed-effects regression:

Table 7 Variation in IS_{SSF} by liquidity quartiles

The values below are the IS_{SSF} for securities by liquidity quartiles, Q1...Q4, where the quartiles have been created based on the median ($IC_{SSF} + IC_s$) for order size $Q = 5000$ for all the securities. The average IC_{SSF} and IC_s for each quartile is reported.

Similarly, the average IS estimates for the stocks in each quartile are presented as a set of three values: the average upper bound (UB), the lower bound LB, and the midpoint of the two MID. This is calculated as the median of the estimate for each stock, and which is averaged across the securities in the quartile.

The table also reports the value of IS-CS ratio to augment inference.

Values marked in boldface indicates that the average IS_{SSF} is significantly different from 0.50 at a 5 percent level of significance.

IC_{SSF}	IC_s	Quartiles by Liquidity	IS_{SSF}			IS-CS ratio
			UB	MID	LB	
0.04	0.04	Q1	0.66	0.61	0.55	1.14
0.06	0.07	Q2	0.63	0.59	0.55	1.13
0.11	0.13	Q3	0.55	0.53	0.51	1.06
0.74	0.29	Q4	0.25	0.24	0.23	0.68

Model 2 : $IS_{SSF,i,t,H/L}$ on two high information dummy variables, $HIGH-INF_j^+$, $HIGH-INF_j^-$ to differentiate between positive and negative news.

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1^+ HIGH-INF_{t,j}^+ + \beta_1^- HIGH-INF_{t,j}^- + \epsilon_{i,t,j}$$

where the dummies takes value 1 for a high information period, H, and 0 otherwise. Here β_1^+ is the coefficient that captures the effect of leverage during periods when information is high and positive, and β_1^- captures the effect of leverage when information is high and negative. If $\hat{\beta}_1^+ \geq \hat{\beta}_1^-$, it weakens the support for the hypothesis that the high observed average IS_{SSF} for the NSE equity markets is because of funding constraints of the trader. If $\hat{\beta}_1^+ < \hat{\beta}_1^-$, then it will strengthen the argument. Model 2 estimations show that the size of the effect is *higher* for bad news and it is significant: $\hat{\beta}_1^-$ is 6.9 percent while $\hat{\beta}_1^+$ is 5.5 percent. This suggests that traders use the SSF more when the information is negative.

Thus, while it is difficult to fully separate the influence of information arrival and that of the funding constraints of traders, the above results suggests that the level and temporal variation of price discovery of the SSF markets on the NSE is the outcome of the response of traders to a combination of information arrivals and their funding constraints.

6.2 Cross-sectional variation in price discovery

Table 2 shows that there is significant cross-sectional variation in the sample estimates of IS_{SSF} . We analyse the price discovery of liquidity quartiles, where liquidity is measured as $IC(Q)$ for a market order of $Q = \$5000$, averaged across IC_{SSF} and IC_s .³¹ The results of this

³¹Since $IS_s = 1 - IS_{SSF}$, we present only the IS_{SSF} estimates in the rest of the paper.

Table 8 Fixed-effects panel regressions for IS_{SSF}

The table reports the estimation for the panel data regressions in Models **3-5**, where:

1. Model 3: $IS_{SSF,i,t} = \alpha_i + \beta_2 IC_{(SSF-S),i,t} + \epsilon_{i,t}$
2. Model 4: $IS_{SSF,i,t} = \alpha_i + \beta_3 \hat{\lambda}_{i,t} + \epsilon_{i,t}$
3. Model 5: $IS_{SSF,i,t} = \alpha_i + \beta_2 IC_{(SSF-S),i,t} + \beta_3 \hat{\lambda}_{i,t} + \epsilon_{i,t}$

The dependent variable is $IS_{SSF,i,t}$. Values in boldface indicate that the coefficients are significant at a 5 percent level using one tail test based on clustered standard errors of the estimated coefficients.

	Model 3		Model 4		Model 5	
	Estimate	Std. Error	Estimate	Std. Error	Estimate	Std. Error
$IC_{(SSF-S),i,t}$	-0.3085	0.0857			-0.3093	0.0853
$\hat{\lambda}_{i,t}$			0.0575	0.0334	0.0618	0.0335
Adjusted R ²	0.22		0.21		0.22	

non-parametric analysis is presented in Table 7. These show that the average IS_{SSF} is greater than 50 percent for liquid securities (Q1 and Q2). This validates the hypothesis H_3 that the SSF market dominates price discovery where there is greater liquidity. In contrast, the IS_{SSF} for the Q4 securities is less than 50 percent, showing that the spot dominates price discovery for those securities that have relatively lower liquidity of the SSF. These results are also consistent with Equation 1 which states that the lower are the transactions costs on the SSF market compared to the spot market (lower $IC(Q)$), the more likely the trader is use the SSF (higher IS_{SSF}).

The parametric approach involves estimating a series of fixed-effects models and examining the coefficients on leverage, $\hat{\lambda}_{i,t}$, and the liquidity difference, $IC_{(SSF-S),i,t}$, as follows:

Model 3 : $IS_{SSF,i,t}$ on the liquidity difference, $IC_{(SSF-S),i,t}$

$$IS_{SSF,i,t} = \alpha_i + \beta_2 IC_{(SSF-S),i,t} + \epsilon_{i,t}$$

Model 4 : $IS_{SSF,i,t}$ on leverage estimates, $\hat{\lambda}_{i,t}$

$$IS_{SSF,i,t} = \alpha_i + \beta_3 \hat{\lambda}_{i,t} + \epsilon_{i,t}$$

Model 5 : $IS_{SSF,i,t}$ on both the liquidity difference, $IC_{(SSF-S),i,t}$, and leverage estimates $\hat{\lambda}_{i,t}$

$$IS_{SSF,i,t} = \alpha_i + \beta_2 IC_{(SSF-S),i,t} + \beta_3 \hat{\lambda}_{i,t} + \epsilon_{i,t}$$

All three models use estimates of average daily IS_{SSF} and leverage, and the observed market liquidity differences from the LOB, for all $i = 1 \dots N$ securities and $t = 1 \dots T$ days in the sample period. An examination of Cook's Distance suggested the presence of influential observations. Hence, the data is winsorized at the 99th and 1st percentiles.

The results presented in Table 8 validate the predictions from Equation 1 for how the trader chooses between the SSF and the spot market. $\hat{\beta}_2$ is negative and significant, which is consistent with the hypothesis that the more negative the liquidity difference between the SSF and the spot market, the higher will be the price discovery of the SSF. $\hat{\beta}_3$ is positive and significant, which is consistent with the prediction that the higher the leverage, the higher will be the price discovery role by the SSF. These results hold for price discovery at the level of individual securities, even when the SSF does not have a dominant role in price discovery for the market as a whole.

What matters more when information arrives?

Finally, we analyse whether the trade-off between leverage and the liquidity differential presented in Equation 1 holds during periods of high information. If the marginal trader uses leverage in order to increase returns from information, then we would expect to see higher values of $\hat{\beta}_3$ during such times. However, if traders use leverage to preserve trading capital, then we expect to see a stronger correlation between IS_{SSF} and the liquidity differential instead.

The following pair of fixed-effects panel regression models are estimated:

$$\textbf{Model 6} : IS_{SSF,i,t,j} = \alpha_i + \beta_1 \text{HIGH-INF}_j + \beta_2 IC_{(SSF-S),i,t,j} + \beta_3 \hat{\lambda}_{i,t,j} + \epsilon_{i,t,j}$$

$$\textbf{Model 7} : IS_{SSF,i,t,j} = \alpha_i + \beta_1^+ \text{HIGH-INF}_{t,j}^+ + \beta_1^- \text{HIGH-INF}_{t,j}^- + \beta_2 IC_{(SSF-S),i,t,j} + \beta_3 \hat{\lambda}_{i,t,j} + \epsilon_{i,t,j}$$

where HIGH-INF_j is the dummy variable for high information periods as defined for the regressions to test Hypothesis H₂ and H₃. The results are presented in Table 9, which shows a highly significant positive estimate for the coefficient of HIGH-INF . The coefficient of liquidity remains negative and significant, which implies that during periods of high information flows, the liquidity difference continues to be important to explain the cross-sectional variation in IS_{SSF} . The coefficient of the leverage remains positive but is insignificant. These results show that the extent of cross-sectional variation in the increase in the IS_{SSF} is related to the intra-day variation in the market liquidity rather than in the leverage.

7 Reproducible research

An R package named `ifrogs` has been released into the public domain, with an open source implementation of the IS calculations used in this paper.³² An example with data in this R package replicates the values of IS, CS and IS-CS ratio for one day for one firm, in Appendix B.

³²Information about this package and some data for replication is available from http://ifrogs.org/releases/ThomasAggarwal2013_priceDiscovery.html

Table 9 Fixed effects panel regressions of intraday variation in IS_{SSF} during high information periods

The table presents regression results for a fixed effects model, **Model 6**:

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1 IC_{(SSF-S),i,t,j} + \beta_2 \hat{\lambda}_{i,t} + \beta_3 HIGH-INF_j + \epsilon_{i,t,j}$$

where $HIGH-INF_j = 1$ for high volatility periods of the trading day, when j = information arrival during the trading day t and 0 for all other periods.

The second fixed effects model, **Model 7**, additionally tests for asymmetric behaviour in the variation in IS_{SSF} depending upon the direction of the information:

$$IS_{SSF,i,t,j} = \alpha_i + \beta_1 IC_{(SSF-S),i,t,j} + \beta_2 \hat{\lambda}_{i,t} + \beta_3^+ HIGH-INF_j^+ + \beta_3^- HIGH-INF_j^- + \epsilon_{i,t,j}$$

where $HIGH-INF_j^+ = 1$ for high information periods with positive returns on Nifty and 0 otherwise. Similarly, $HIGH-INF_j^- = 1$ for high information periods with negative returns on Nifty and 0 otherwise. Values marked in boldface indicate that the coefficients are significant at 5 percent, based on clustered standard errors.

	Model 6		Model 7	
	Estimate	Std. Error	Estimate	Std. Error
$IC_{(SSF-S),i,t,j}$	-0.770	0.076	-0.769	0.075
$\hat{\lambda}_{i,t}$	0.024	0.033	0.027	0.033
HIGH-INF	0.055	0.005		
HIGH-INF ⁺			0.049	0.006
HIGH-INF ⁻			0.061	0.006
Adjusted R ²	0.21		0.21	

8 Conclusion

The literature on price discovery of single stock derivatives presents a puzzle because the empirical results show that they play a weak role in price discovery. This contradicts our prior that if traders prefer leverage to trade information (Black, 1975), these derivatives will dominate price discovery.

In this paper, we offer a possible resolution of this puzzle. The resolution is partly rooted in better measurement which avoids drawbacks in the existing literature. These include problems of estimation risk involved in using implied prices from single stock options markets that most of the empirical literature studies; problems of analysing price discovery when trading is fragmented across several derivatives markets while the empirical literature focusses only a few; problems of reporting inference based on averages which can mask changes in price discovery in response to information arrivals, which is sporadic, especially for single stocks. While it is well known that IS measurement works better with very high frequency data, few papers have worked with data at one second frequency.

Another part of the resolution lies in thinking about the funding constraints of financial market intermediaries: that traders use leverage as a response to liquidity constraints. Brunnermeier and Pedersen (2009) suggests that, in equilibrium, traders with funding constraints will chose to invest in low margin securities to preserve capital efficiency. By analogy, traders with funding constraints could similarly use single stock futures.

Unlike most global markets, the NSE equity market is a unique situation: it has trading systems that yield perfectly synchronised data for prices and liquidity from both the SSF and the spot. In both markets, NSE has dominant market share, so the confounding effects of fragmented price discovery across many trading venues are absent. It also has microstructure characteristics, and is in a regulatory environment, that imposes relatively more stringent funding constraints on traders.

We find that the price discovery of the stock futures on the NSE is 49 percent, which rises to 55 percent during high information periods. Thus, this paper is the first to report that stock futures can *dominate* price discovery. Further, we find that the IS is 61 percent when the news is negative, which shows a higher preference for SSF when informed traders need to short sell. Since the cost of borrowing securities is higher on the NSE because of short-selling restrictions, we infer that traders make their choices not just in response to information flows but also to preserve their trading capital.

We develop a partial equilibrium framework to understand the trade-off between leverage and pre-trade liquidity available on the two markets, and find that the empirical analysis supports predictions from this analysis. Compared with the simple averages of the existing literature, this analysis reveals complexity in the behaviour of price discovery of SSF contracts, both within the day and across securities, as an outcome of the choices of traders. While the fixed effects panel regressions are not causal, they are immune to the endogeneity bias induced by omitted variables.

We do not have sufficient information about the traders in the market to make an causal link between the higher price discovery on the NSE with the funding constraints of traders at the NSE. However, there are two reasons that we suggest that it is a contributory factor. The first is that the price discovery of the futures is high on average, which cannot be attributed entirely to information arrival particularly for a single stock where information arrivals tend to be sporadic and far less frequent as compared to market wide news. On the other hand, funding constraints are present at all times, and can influence the average price discovery. The second is that the SSF dominate the price discovery even more strongly during the arrival of bad news. Constraints on short-selling the spot implies that the trader has a stronger preference to use SSF during such times, and this appears as higher price discovery in the SSF.

This analysis opens up several questions for downstream research. If funding constraints influence trader's choices, then the next step can be to develop models that incorporate these constraints. The analytical framework developed in this paper takes liquidity as exogenously given. The empirical validation of the predictions of the model suggests that these simplifications are useful. However, in equilibrium, market liquidity is endogenously determined by the decisions of all traders. This suggests the possibility of a more complex model, where traders have varying degrees of funding constraints and access to information, jointly determining the liquidity in the limit order book. Such a model would help more fully explain why the SSF of some securities have better liquidity and price discovery while others in the same market structure do not.

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A Analysing a marginal trader's choice between the SSF and spot market

Consider the trader's choice between the SSF and the spot market. A security has price P_0 at time $T = 0$, that can either move up to uP_0 with probability p , or down to dP_0 with probability $(1 - p)$ at $T = 1$. This security trades on both the spot and the SSF markets. When information arrives, the trader chooses to trade either the SSF or the spot. We restrict ourselves to a partial equilibrium framework and analyse the decision of an infinitesimally small trader who can treat liquidity and leverage as exogenously determined.

The trader has the choice of either placing an order of value $\$Q$ in the spot market or an order of value λQ in the SSF market, where λ is the leverage of the SSF market. The cost of funding will be the same for both these positions, r_f . As an example, if $\lambda = 5$, and the security has a price of $\$100$, the trader can either invest $\$100$ into the spot market ($Q_s = 100$) or take a position worth $\$500$ ($Q_{\text{SSF}} = \lambda \times Q_s$) in the SSF.

Both the SSF and the spot trade on an open electronic limit order book (OELOB) market, which enables precise observation of the transaction cost that will be incurred prior to order placement. The trader only places market orders. Since the limit order book is visible, the trader knows the cost associated with the spot market $c_s(Q)$ and the futures market $c_{\text{SSF}}(Q)$ for all values of Q . For example, if the mid-quote price is $\$100$, and if a market buy order of value $\$100,000$ is executed at a buy price of $\$101$ per share, then $c_s(100,000) = 0.01$ or one percent of the value of the transaction. We assume that no-arbitrage holds in the relationship between the spot and the SSF market, and there are no dividend payments. This ensures that a simple cash-and-carry model holds. Lastly, we represent the preferences of the trader by assuming that he maximises the ratio of expected return per unit risk of the transaction, $[E(r_M)/\sigma_M]$, where M is either "S" for the spot market or "SSF" for the single stock futures market.

The spot market position of Q will incur funding cost of r_f and trading cost $c_s(Q)$. Then,

$$\begin{aligned} E(r_s(Q)) &= p[u - c_s(Q) - r_f] + (1 - p)[d - c_s(Q) - r_f] \\ &= p[u - d] + d - c_s(Q) - r_f \\ &= E(r) - c_s(Q) - r_f \end{aligned}$$

Where $E(r)$ is the inherent return on the security in the spot market for a position of size Q . For the same cost of funding r_f , the trader can also take a position of size λQ on the SSF market which has a related trading cost of $c_{\text{SSF}}(\lambda Q)$.

$$\begin{aligned} E(r_{\text{SSF}}(\lambda Q)) &= p[\lambda u - c_{\text{SSF}}(\lambda Q) - r_f] + (1 - p)[\lambda d - c_{\text{SSF}}(\lambda Q) - r_f] \\ &= \lambda(p[u - d] + d) - c_{\text{SSF}}(\lambda Q) - r_f \\ &= \lambda E(r) - c_{\text{SSF}}(\lambda Q) - r_f \end{aligned}$$

In the calculation of the risk of the two positions, we note that the payoff for both the spot and the SSF is linear but that the slope of the SSF market payoff is higher by the amount of the leverage, λ .³³ This makes the SSF a riskier proposition than the spot for speculators on the same base of capital deployed.

$$\begin{aligned}
\sigma_s^2(Q) &= E(r_s^2) - [E(r_s)]^2 \\
&= p[u - c_s(Q) - r_f]^2 + (1-p)[d - c_s(Q) - r_f]^2 - [E(r_s)]^2 \\
&= p(1-p)[u - d]^2 \\
\sigma_s^2(Q) &= \sigma^2 \\
\sigma_{\text{SSF}}^2(\lambda Q)^2 &= E(r_{\text{SSF}}^2) - [E(r_{\text{SSF}})]^2 \\
&= p[\lambda u - c_{\text{SSF}}(\lambda Q) - r_f]^2 + (1-p)[\lambda d - c_{\text{SSF}}(\lambda Q) - r_f]^2 - [E(r_{\text{SSF}})]^2 \\
&= p(1-p)[\lambda(u - d)]^2 \\
\sigma_{\text{SSF}}^2(\lambda Q) &= \lambda^2 \sigma^2 \implies \\
\sigma_{\text{SSF}}^2(Q) &= \sigma^2 = \sigma_s^2(Q)
\end{aligned}$$

Where σ^2 is the inherent risk on the security in the spot market.

We now turn to the problem of the trader's choice between the two markets. For a fair comparison, we assume that the trader will choose to place a market order of the same size on both markets, which we set as λQ for the sake of convenient exposition. The transactions costs for a position of λQ on the spot market will be $c_s(\lambda Q)$, and the funding cost will be λr_f . The transaction cost for the same order size on the SSF market will be $c_{\text{SSF}}(\lambda Q)$ and the funding cost will be r_f . The return to risk associated with the trader's choices are:

$$\begin{aligned}
\text{Spot: } \frac{E(r_s(\lambda Q))}{\sigma_s(\lambda Q)} &= \left[\frac{\lambda E(r) - c_s(\lambda Q) - \lambda r_f}{\lambda \sigma_s} \right] \\
\text{SSF: } \frac{E(r_{\text{SSF}}(\lambda Q))}{\sigma_{\text{SSF}}(\lambda Q)} &= \left[\frac{\lambda E(r) - c_{\text{SSF}}(\lambda Q) - r_f}{\lambda \sigma_s} \right]
\end{aligned}$$

The SSF market will be preferred if $E(r_{\text{SSF}}(\lambda Q))/\sigma_{\text{SSF}}(\lambda Q) > [E(r_s(\lambda Q))/\sigma_s(\lambda Q)]$. After removing the common terms, we find that the condition under which the trader will prefer to trade on the SSF market rather than spot market holds is:

$$(\lambda - 1)r_f > [c_{\text{SSF}}(\lambda Q) - c_s(\lambda Q)]$$

B The IS-CS ratio

As described in Section 4, Yan and Zivot (2010) propose a structural representation of the IS and CS measures. Based on the structural parameters, they show that in a two-market

³³Since we assume there is sufficient arbitrage activity to eliminate basis risk in the current setting, it does not appear as a factor of risk.

framework, the reduced form forecasting errors ($e_{i,t}$) can be written as a function of structural shocks ($\eta_{i,t}$) as follows:

$$\begin{bmatrix} e_{1,t} \\ e_{2,t} \end{bmatrix} = \begin{bmatrix} d_{0,1}^P & d_{0,1}^T \\ d_{0,2}^P & d_{0,2}^T \end{bmatrix} \times \begin{bmatrix} \eta_t^P \\ \eta_t^T \end{bmatrix}$$

where,

- item i represents the i^{th} market,
- $d_{0,i}^P$ represents the contemporaneous response of i^{th} market to permanent shocks, and
- $d_{0,i}^T$ represents the contemporaneous response of i^{th} market to transitory shocks.

The paper shows that the Component Share of a market can be written as:

$$CS_1 = \frac{d_{0,2}^T}{d_{0,2}^T - d_{0,1}^T}; \quad CS_2 = \frac{-d_{0,1}^T}{d_{0,2}^T - d_{0,1}^T}$$

The above structural representation indicates that the CS of a market only captures the relative strength of a market's in response to the transitory shock. Thus, the CS for Market 1 will be higher than that for Market 2 if Market 2 responds more strongly to transitory shocks.

Under the assumption that e_t is uncorrelated, Yan and Zivot (2010) show that a structural representation of IS can have three cases. In the first two cases, the IS will correctly capture “who moves first” with respect to new information. In the third case, they showed that the IS of a market can be written as:

$$IS_1 = \frac{d_{0,1}^P d_{0,2}^T}{d_{0,1}^P d_{0,2}^T - d_{0,1}^T d_{0,2}^P}; \quad IS_2 = \frac{-d_{0,1}^T d_{0,2}^P}{d_{0,1}^P d_{0,2}^T - d_{0,1}^T d_{0,2}^P}$$

These show that the IS estimate will capture the contemporaneous response not only to the permanent shocks but also to transitory shocks. In particular, high values of IS for Market 1 could be due only to a strong response of Market 1 to new information but also to a strong response of Market 2 to frictions.

To isolate the effect of the permanent shocks, first, the ratio of IS of the two markets in terms of structural parameters is written as follows:

$$\frac{|IS_1|}{|IS_2|} = \frac{|d_{0,1}^P \times d_{0,2}^T|}{|d_{0,2}^P \times -d_{0,1}^T|}$$

The term $\frac{|d_{0,1}^P|}{|d_{0,2}^P|}$ indicates the relative strength of the Market 1 response to the contemporaneous permanent shock compared to the Market 2 response. If this value is greater than one,

it will indicate that Market 1 reacts more to the permanent shocks in comparison to Market 2. $\frac{|d_{0,2}^T|}{|-d_{0,1}^T|}$ indicates how strongly Market 2 responds to transitory shocks compared to Market 1.

This ratio will be greater than 1 if either of the following holds:

- **Case 1:** $\frac{|d_{0,1}^P|}{|d_{0,2}^P|} > 1$ and $\frac{|d_{0,2}^T|}{|-d_{0,1}^T|} > 1$
- **Case 2:** $\frac{|d_{0,1}^P|}{|d_{0,2}^P|} < 1$ and $\frac{|d_{0,2}^T|}{|-d_{0,1}^T|} > 1$
- **Case 3:** $\frac{|d_{0,1}^P|}{|d_{0,2}^P|} > 1$ and $\frac{|d_{0,2}^T|}{|-d_{0,1}^T|} < 1$

If $\frac{|IS_1|}{|IS_2|} > 1$ holds along with **Case 2**, it will imply that the high IS value of Market 1 is not due to greater reaction of Market 1 to permanent shocks, but due to Market 2's strong response to transitory shocks. Market 1 will be called to dominate the price discovery process if and only if both $\frac{|IS_1|}{|IS_2|} > 1$, as well as $\frac{|d_{0,1}^P|}{|d_{0,2}^P|} > 1$. Thus, if we can measure $\frac{|d_{0,1}^P|}{|d_{0,2}^P|}$, we can identify which market dominates the price discovery process.

$\frac{|d_{0,1}^P|}{|d_{0,2}^P|}$ can be measured using the ratio proposed by Yan and Zivot (2010) in the following manner:

$$\frac{|IS_1 \times CS_2|}{|IS_2 \times CS_1|} = \frac{|d_{0,1}^P|}{|d_{0,2}^P|}$$

As an example of the calculation of IS-CS ratio, we calculate the IS and CS values for one firm (RELIANCE) in the sample for one day (6th May 2009). The resulting IS, CS and IS-CS ratio are presented in Table 10.

Table 10 IS, CS and IS-CS ratio estimates using prices on 6th May 2009 for RELIANCE

The table shows the IS and CS values generated for RELIANCE using one second spot and the futures midquote prices of May 6, 2009.

IS_{SSF}^U shows the upper bound of the information share for the SSF market, while IS_{SSF}^L shows the lower bound. IS_{SSF} shows the average of the two. CS_{SSF} indicates the component share weights for the SSF market. IS-CS ratio is calculated using the average IS and CS values.

In this example, both $IS_{SSF} > 0.5$ and the IS-CS ratio > 1.0 . This implies that the SSF prices are responding faster than spot prices, and the response is to a permanent rather than a transitory shock.

IS_{SSF}^U	IS_{SSF}^L	IS_{SSF}	CS_{SSF}	IS-CS ratio
0.98	0.93	0.96	0.89	2.96

C Estimating leverage

Since the daily leverage values are not archived by the NSE, we require an empirical strategy to estimate a time series for leverage for all the securities in the sample for the period of the analysis. The empirical strategy we employ to obtain $\hat{\lambda}_{i,t}$ for day t of security i is as follows:

1. Obtain a sample of one month of daily leverage values directly from the NSE.
2. Alongside, calculate the daily average σ_s as the *realised volatility* (RV) for the security.³⁴
3. Estimate a linear relationship between the daily leverage and daily σ_s using the following fixed-effects regression:

$$\lambda_{i,t} = \gamma_{1,i} + \gamma_2 \sigma_{s,i,t-1} + \gamma_3 \sigma_{s,i,t-1}^2 + \epsilon_{i,t}$$

where $\lambda_{i,t}$ denotes the leverage on security i at time t for the month of June, which was obtained through a request from the NSE. σ_s denotes the volatility of security i at time $t-1$ for that period.

4. Use the above equation with the daily $\sigma_{i,(t-1)}$ measured from the price data to estimate the daily $\hat{\lambda}_{i,t}$ for the security i , subject to a maximum value of 8.

At the end of the estimation procedure, we find that the following regression offers a good approach to obtain leverage estimates for all the securities in the sample.

$$\hat{\lambda}_{i,t} = \hat{\gamma}_{1,i} - 0.59 \sigma_{s,i,t-1} + 0.15 \sigma_{s,i,t-1}^2$$

where $\hat{\gamma}_{1,i}$ denotes the intercept estimate obtained for each security from the fixed-effects regression. The in-sample adjusted R^2 of the regression is 99%. We then use estimates of the daily values of σ_s calculated using the realised volatility approach for a daily time series of $\hat{\lambda}_{i,t}$ for each security.

³⁴Here, RV is the sum of intraday squared returns, calculated at a five-minute frequency and then annualised.